

MME305 Measurements & Instrumentation Calibration Sheet
Pancake Project, Load / Torque Cells

LOAD CELL CALIBRATION SHEET

Instrumentation

Item	Details
Instrument Type	Load Cell
Rated Capacity	2 kg
Manufacturer	Uxcell
Model	1034748
Serial Number	D1
Calibration Date	04/08/26
Location	GAR 145, Miami University, 650 E High St. Oxford Ohio
Environmental Conditions	Temp: RT
DAQ Used	NI USB 6001
Excitation Voltage	5V

Weights Used

Reference	1	2	3	4	5
Weights					
Weight (kg)	0.02	0.05	0.1	0.20	0.5

Calibration Method

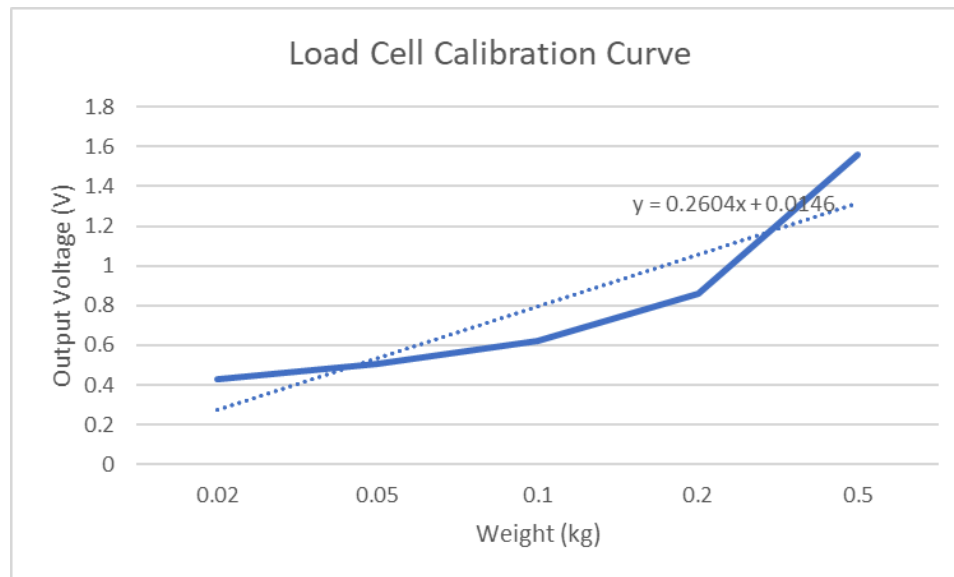
Method: Weights of 5 known masses were placed one at a time upon the center of the load cell in no particular order. The voltage output was read on the DAQ for each weight and then recorded in an excel document. Using this data, a plot was constructed and trendline was formed to give a

calibration constant (the slope of the curve) and a zero load offset which was displayed as the y intercept.

Calibration Results

Parameter	Tolerance	Result
Calibration Constant	NA	0.2604 V/kg
Zero Drift	± 0.05%FS	0.0146 V

Calibration Curve



Signatures

Technician Signature: Kyle Buttray Date: 4/08/26

Technician Signature: Troy Jones Date: 4/08/26